Physics lab

1. Determination of Dielectric Constant and Phase Transition Temperature of Dielectric Materials

2. To study the I-V Characteristics of P-N Junction Diode and Resistance Evaluation

3. Find the Electrical Conductivity and Energy Gap of Germanium (Ge) Crystal

4. Study Hall Effect in Semiconductors & find Hall Coefficients. Hall Voltage, and Conductivity

5. Study Characteristics of Thermistor: Determination of Constants A and B

6. Draw Hysteresis Loop for Ferromagnetic Material (B-H Curve)
7. Study V-1 Characteristics of Solar Cell: Fill Factor and Series Resistance Calculation 8. Visualization Energy Levels of 1-Dimensional Potential Box Using Schrödinger Wave Equation in Python

9. Visualization of Allowed Energy Levels and Kronig-Penney Model in Python/MATLAB

10. Determine the Density and Elastic Properties of Oxide Glasses/polymers Using Machine learning algorithms

11. Calculate the Numerical Aperture (NA) and Acceptance Angle of Optical Fiber

12. Find the Wavelength of Laser Source using diffraction grating
13. To study the Transition Temperature Measurement of High-Temperature Superconducto

14. Find the Rigidity Modulus of Wire Using Torsional Pendulum

15. To estimate the pressure using optical fiber sensor.